IAE PARIS — SORBONNE BUSINESS SCHOOL · IDGP ACADEMY

Digital Tool Appraisal

The instrument for the afternoon's group activity. One tool, about twenty minutes, filled in by the people who will live with the decision.

17 October 2026 · 14:00–17:00

Prof. Dr. Thiago Lima

Companion to the Digital Tools & Benchmarking reading notes

This printed version is for working in the room. The same instrument is fillable and exportable online, which is the version to use when the appraisal has to leave the session.

Contents

Does the tool change what the organisation owns or what it is able to do; where it contains AI, what the system was told to want, to know and to see, and at which link the chain from a working model to a result breaks; and which capability gaps to close first.

ONE **Why a tool can be excellent and still return nothing** 4

TWO **The instrument** 7

THREE **Reading the result** 17

FOUR **Sources** 18

> **There is no score.** The instrument restates your own answers and locates the gap. A composite figure would hide the only thing worth knowing, which is where the chain stops.

Four questions decide whether a digital investment returns anything. Does it change what the organisation *is able to do*, or only what it *owns*? Where it contains AI: what has the system been told to want, to know, and to see — and at which link does the chain from a working model to a result actually break? And what is the reference already doing that you are not: the routine, not the number?

This instrument asks all four, and then turns the answers outward: a capability maturity read of the organisation rather than of the tool, and a six-month roadmap across the three gaps that most constrain performance. It takes about twenty minutes for one tool, and it is designed to be filled in by the people who will live with the decision rather than by the people selling it.

ONE

Why a tool can be excellent and still return nothing

The afternoon session establishes the frame. Value is not the sum of what an organisation owns and how it works — it is that sum scaled by its capacity to improve.

$$V = C \times (A + P)$$

assets and processes are additive and purchasable · capability is a coefficient

A platform is almost always an **A** decision, sometimes a **P** decision, and very rarely a **C** decision. That is not a criticism of platforms — it is a statement of what they are. The error is paying for A and booking the benefit as though C had moved.

Technical debt accumulates in A and P. Philosophical debt accumulates where C should be.

Schrage and Kiron argue that the binding constraint on value from AI is not technical but philosophical: organisations invest heavily in algorithms, data infrastructure and talent while never articulating what their systems should value, how they should know, and what they should represent about reality. The unarticulated assumptions do not stay harmless. They compound.

The parallel with technical debt is exact, and so is the difference. Technical debt is a shortcut in the artefact. **Philosophical debt is an absence in the organisation** — nobody wrote down the hierarchy of purposes, so nobody can audit it, so nobody can improve it. An organisation that cannot say what its system is optimising for has no basis on which to run an improvement routine over it. That is a deficit in C, arriving dressed as a deficit in A.

The three pillars, and where you have already met them

Each of Schrage and Kiron's pillars restates something the day has already established in another vocabulary. That is the reason to take them seriously rather than as a separate topic.

Pillar	What it governs	Where it appears in the session
Teleology	What the system should achieve — the hierarchy of purposes, and what happens when objectives conflict	Order winners and qualifiers. A system optimising the wrong metric is one whose purposes were never ranked — the same error as over-investing in a qualifier.

Epistemology	What counts as valid knowledge — how the system acquires it, validates it and expresses certainty	The digital andon. A system that knows when it does not know, and escalates, *is* signalling at the source rather than reporting on a cycle.
Ontology	How the system represents reality — which entities it recognises, which relations it models, which variables it ignores	Causal ambiguity. Representing a supplier by price and lead time alone is the same error as benchmarking the indicator instead of the practice that produces it.

> **THE FAILURE THESE PREDICT**
>
> A sourcing system optimised for cost and delivery that, two years on, has systematically excluded suppliers in regions the firm's own strategy prioritised. The fault is not in the model. It is that the hierarchy of objectives was never written down, so the trade-off was resolved by default — and silently, because nothing in the architecture was built to surface it.

Why this matters more for AI than for anything else you buy

Every tool embeds assumptions. What distinguishes an AI system is that it **acts on them at scale, continuously, and without re-stating them**. A spreadsheet with a wrong assumption produces one wrong number that somebody reads. A model with an unarticulated teleology produces ten thousand decisions a month, each technically correct and strategically wrong, and reports high confidence throughout.

This sharpens as systems become agentic. A predictive system recommends and a human decides; an agentic system formulates plans and acts. At that point the hierarchy of purposes stops being documentation and becomes the control surface — the only place where the boundary of autonomy is actually set.

And what the measured evidence says

The pillars are an argument about what goes unarticulated. There is also a measurement, and it comes from a systematic review of the implementations that actually happened. Of 1,271 records screened, 37 had both an implementation and a comparison — roughly one for every eight that end at a model or a simulation. In the studies with no effect, predictive performance was high and stable. **Model accuracy does not separate success from failure.** What separates them is a chain of six conditions, each necessary and none sufficient.

0
Evidence base
THE TOOL

1
Technical capability in operation
THE TOOL

2
Workflow integration
CAPABILITY 1

3
Translation into action
CAPABILITY 2

4
Organisational outcome
THE RESULT

5
Net value
THE RESULT

The two middle conditions are not properties of the tool at all. They are capabilities of the organisation that bought it: whether the output reaches whoever decides while the decision is still open, and whether anyone is able to act on it with the resources that action requires. That is C, in a form you can measure — and it is where the reviewed evidence says value is most often lost.

> **PROVENANCE, AND THE LIMIT OF THE EVIDENCE**
>
> The chain is drawn from a systematic review of AI implementations conducted under PRISMA 2020 (Lima, manuscript in preparation). **Its evidence base is hospital implementations.** The conditions are stated here in general organisational terms, which is an extension and not a finding of the review. Use it as a structure for appraisal, not as a citable result.

TWO

The instrument

Appraise one tool at a time. Answer from what the organisation can actually evidence, not from what the business case promised. **"Not documented" and "nobody knows" are the most useful answers here** — they are precisely what the instrument is looking for.

Digital Tool Appraisal

Sections 1, 2 and 7 take about ten minutes; with the AI sections, about twenty-five. Sections 5 and 6 are usually completed after the session.

1 · The tool

Tool or platform

What it is called

The workflow it automates

Be specific. "Improves collaboration" is not a workflow.

One line

Where it stands

- ☐ Under evaluation — not yet bought
- ☐ Bought, being implemented
- ☐ In production under a year
- ☐ In production more than a year

Does it include AI or machine learning?

Anything that learns from data, scores, ranks, forecasts, classifies or generates. If a vendor calls it AI and you are not sure what it does, answer yes — that uncertainty is itself a finding.

- ☐ Yes
- ☐ Unsure — the vendor says so; we cannot say what it does
- ☐ No

2 · VCAP triage — what does it actually change?

Which term does it move?

Training people to operate the new system is part of delivering the asset. C means the organisation ends up better at improving its processes than it was before.

- [ ] A — we own something we did not own before
- [ ] A and P — the way work flows changes too
- [ ] A, P and C — we will also be better at improving afterwards
- [ ] Honestly, none of them yet

If you claimed C: what will the organisation be able to do afterwards that it cannot do now — and that does not depend on this system?

Leave blank if you did not claim C. A blank here is a legitimate and common answer.

Does it limit work in progress?

Work in progress is everything started and not yet finished. With the same capacity, more in flight means each item waits longer — Little's Law from the morning. A tool that makes work easier to *start* without capping what is in progress lengthens lead time by construction.

- [ ] Yes — there is an explicit cap, and it is enforced
- [ ] There is a cap that nobody enforces
- [ ] No — it only makes starting work easier

Where will each fact live afterwards?

Every type of fact — a schedule date, a risk, a cost figure — needs exactly one authoritative home, with everything else reading from it. The moment the same fact lives in two systems, somebody reconciles them forever and the dashboards become negotiations.

- [ ] It replaces existing sources — one authoritative home afterwards
- [ ] It adds another place where the same fact lives
- [ ] It consumes from an existing authoritative source

Adoption — expected against actual

Of the people meant to do their work *inside* it — not just log in — what share actually do? Write "too early" if it is not live. A tool used by 40% of its intended users does not deliver 40% of a single-source benefit; it delivers close to none, because the other 60% still has to be reconciled.

e.g. expected 100%, actual about 40%

How long before a competitor with the same budget has the same thing?

This is the test for whether the tool buys advantage or only parity. Anything purchasable is available to a rival at roughly the same price.

- ☐ Months — they would buy the same product
- ☐ A year or two
- ☐ Longer, because of what we would also have to build

> **The next two sections apply only where the tool contains AI or machine learning.** If the last question of section 1 is answered no, go straight to section 5. In the online version these sections appear and disappear on their own.

3 · Philosophical debt audit

Three pillars, after Schrage and Kiron. The question in each case is not whether the system is good, but whether the organisation has *articulated* the assumption — because an assumption nobody wrote down is one nobody can audit or improve.

Teleology — what it is told to want

The hierarchy of purposes. Most critical, most often skipped.

What is it optimising for, precisely?

The number the system actually tries to maximise or minimise — cost per order, forecast error, time to resolution — not the business goal it was sold against. If nobody can say, write "unknown". That is the finding, not a gap in your answer.

The metric it actually maximises or minimises

Is the hierarchy of objectives documented?

When cost, service level, risk and sustainability conflict, is it written down which wins?

- ☐ Yes — written, owned and reviewed
- ☐ Written once, never revisited
- ☐ Not documented — it is implicit in the configuration
- ☐ Nobody in this organisation could answer it

What has it already sacrificed to hit its metric?

Every optimiser trades something away. If you cannot name what, that is the finding.

Epistemology — what it is allowed to count as knowing

How it acquires knowledge, validates it, and expresses certainty.

Does it express uncertainty, and does anything happen when it is high?

- [ ] Yes — low confidence escalates to a human, automatically
- [ ] It reports confidence; nothing is triggered by it
- [ ] It returns an answer with no stated confidence
- [ ] We do not know

What would invalidate its training data?

A model learns the conditions that generated its data. Name the event that would end those conditions — a market shift, a regulation, a reorganisation, a pandemic.

Correlation or causation?

Example: the model flags suppliers with late deliveries as risky. Is lateness the cause of the trouble, or do both come from something else — a region, a contract type, a season? If nobody has checked, the system acts on a correlation, and it will keep acting on it after the underlying cause has changed.

- [ ] Tested — we know which relationships are causal
- [ ] Partly — for the major drivers only
- [ ] It acts on correlations; nobody has tested them

Ontology — what it is able to see

Which entities it recognises, which relations it models, which variables do not exist for it.

Which things, and which attributes of them, does it see?

A system only reasons about what it has a field for. List the objects it knows — suppliers, orders, tickets, people — and the handful of attributes it uses to describe each.

e.g. suppliers by price, lead time and defect rate

Name one variable that matters to this decision and that the system cannot see.

Geopolitical exposure, a relationship, a regulatory change, a supplier's own fragility. If nothing comes to mind in sixty seconds, that is worth noting as an answer.

Who reviews what the system represents, and when?

- [ ] A named owner, on a fixed cadence
- [ ] Only when something goes wrong
- [ ] Nobody — it has the representation it was built with

4 · Capability maturity — the six conditions

The chain from a working model to an organisational result, after Lima (manuscript in preparation). Each condition is necessary and none is sufficient, so the useful answer is the *first* one that is not a clear yes. **Not measured** is a legitimate and common answer — and a more tractable finding than a failure.

Read across, the six answers *are* a maturity statement, and a more defensible one than a stage number: they say which conditions this organisation can currently satisfy and demonstrate, which it satisfies without evidence, and which it has never observed. There is no composite score here on purpose. A single figure would hide the only thing worth knowing, which is *where* the chain stops.

0 · Evidence base — has this kind of application produced a measured result in a real organisation?

"Kind of application" means the job: demand forecasting, scheduling, document generation, risk scoring, case prioritisation. Not a model, not a simulation, not a pilot that reported accuracy — an implementation somebody measured afterwards, ideally in an organisation like yours.

- [ ] Yes, in a comparable organisation
- [ ] Partly — results exist but diverge
- [ ] No, only modelling or vendor material
- [ ] Not measured, or nobody has looked

1 · Technical capability in operation — does it still perform on *our* data, the way we actually run it?

Vendor accuracy was measured on the vendor's data, with the vendor's settings. Has anyone checked it on ours, after go-live, with the cut-off we actually use for a flag or an alert? And count only the outputs that tell people something they did not already know — a warning that arrives after everyone has noticed is accurate and useless.

- [ ] Yes, verified locally after deployment

- [ ] Partly — verified once, not monitored
- [ ] No, or performance has drifted
- [ ] Not measured

2 · Workflow integration — does the output reach whoever decides, while the decision is still open?

The first of the two organisational conditions. Name the person and the step, not the dashboard.

- [ ] Yes — named recipient, inside the decision window
- [ ] Partly — it arrives, but late or to the wrong role
- [ ] No — it is available rather than delivered
- [ ] Not measured — we do not know if it is used

Who receives the output, at which step, and is it in fact used?

Two thirds of staff rarely or never opening the tool is the commonest observed failure, and it is invisible unless somebody asks.

Role · step in the process · evidence of use

3 · Translation into action — is there an owner able to act, with the resources the action requires?

The second organisational condition. If acting on the output needs three approvals, a budget line and a resource nobody has, an earlier warning changes nothing — the constraint was never information.

- [ ] Yes — named owner, resourced, low barrier
- [ ] Partly — owner named, resources contested
- [ ] No — identification without capacity to act
- [ ] Not measured — we do not know the follow-through rate

The action, its owner, and what it depends on

Write the sentence the business case should contain: when the output says X, *who* does *what*, using *what*.

4 · Organisational outcome — which indicator should move, measured how?

One named indicator, one data source, one baseline — and not a satisfaction survey. Where perception and objective measurement were collected together, they diverged.

- [ ] Yes — named indicator, source and baseline
- [ ] Partly — indicator named, no baseline
- [ ] No — benefit stated in general terms

- [ ] Measured only by what users report

5 · Net value — what does the gain cost, and who bears it?

Who is de-prioritised, which work moves to other people or other hours, and what checking the output takes. The hour saved in the back office that reappears as an hour of overtime in the front office is not a gain; it is a relocation.

- [ ] Yes — the displaced cost is identified and accepted
- [ ] Partly — suspected but not quantified
- [ ] No — only the gain has been counted
- [ ] Not measured

How good is the evidence you just used to answer?

Three things make a result trustworthy: a comparison group measured at the same time, an objective measure rather than what people said they felt, and a source that is not selling the tool. Tick the weakest point in what you relied on. Vendor material usually fails at least two of the three.

- [ ] None — a comparison group at the same time, an objective measure, an independent source
- [ ] No comparison group — only before-and-after, so anything else that changed gets credited to the tool
- [ ] Measured by what users reported, not by an objective figure
- [ ] The figures come from the vendor
- [ ] More than one of these

5 · The reference

Somebody already does this well. The question is not what their number is, but what routine produces it — and what you would have to accumulate before that routine would work here.

Who, or what standard?

The most useful reference is often outside your industry. Xerox fixed its warehouse by studying a catalogue retailer, not a competitor.

An organisation, or a published standard

What does the reference do that you do not?

The practice, not the indicator. "They are 30% faster" is a number; "they review the representation quarterly with a named owner" is a practice.

Would you be copying an asset, a process, or a rule-in-use?

- [ ] A — an asset we would buy
- [ ] P — a process we would document and copy
- [ ] C — a way of working that lives in people's habits, which we would have to build

What would have to be true here for that practice to work?

What would our people need to already know, or already be able to do, for this to take here? Practices fail to transfer for reasons of knowledge far more often than of motivation — including between two units of the same company.

How long did the reference take to build it?

- [ ] Months
- [ ] A few years
- [ ] A decade or more
- [ ] Unknown — nobody here knows

Adopt, adapt or discard

- [ ] Adopt — conditions are already in place
- [ ] Adapt — translate it to our context
- [ ] Discard — it depends on conditions we do not have and cannot reasonably accumulate

6 · Capability gaps and the six-month roadmap

The appraisal above concerns one tool. A roadmap concerns the organisation. Name the three capabilities whose absence most constrains performance — capabilities, not products: "a single authoritative home for schedule dates" rather than "a new PPM system" — then place each one on the two axes that decide sequence. **Strategic value**: how much it moves something customers choose you for, or removes a constraint that keeps recurring. **Feasibility**: whether the conditions to build it exist today — the people, the data, and the authority to change the process.

Gap 1

The capability that is missing, in one line

Gap 1 — strategic value, feasibility, and when

- [ ] High
- [ ] Medium
- [ ] Low

Gap 2

The capability that is missing, in one line

Gap 2 — strategic value, feasibility, and when

- [ ] High
- [ ] Medium
- [ ] Low

Gap 3

The capability that is missing, in one line

Gap 3 — strategic value, feasibility, and when

- [ ] High
- [ ] Medium
- [ ] Low

What has to be true by the end of month two?

One sentence, measurable. A roadmap whose first checkpoint cannot be stated is a list of intentions with dates attached.

7 · The verdict

Where is the largest debt?

- [ ] Technical — the artefact itself
- [ ] Philosophical — unarticulated purpose, knowledge or representation
- [ ] Adoption — it works and people do not use it

- [ ] Capability — we would not be able to improve it after buying it
- [ ] None that we can see

The decision

- [ ] Proceed
- [ ] Proceed once named conditions are met — write the first of them as the step below
- [ ] Pause — the debt has to be paid first
- [ ] Do not proceed

The first step, and who owns it

One line

THREE

Reading the result

The instrument has no score, deliberately. It produces three things worth looking at.

- **Which term you could defend.** Most honest appraisals land on A, or on A and P. That is not a failure — it is a correctly identified purchase. The failure is claiming C and being unable to say, in one sentence, what the organisation will be able to do afterwards without the system.
- **How many pillars are unarticulated.** Not how good the system is — whether anyone wrote the assumption down. A pillar nobody has articulated cannot be audited, and therefore cannot be improved. Three unarticulated pillars in a system that is already acting autonomously is the condition under which the sourcing failure above becomes likely rather than unlucky.
- **Whether the debt is payable before or only after.** Teleology is cheap to articulate before deployment and expensive afterwards, because by then the system has been acting on the implicit hierarchy and the organisation has accumulated two years of decisions made under it.

> **THE ONE QUESTION TO TAKE INTO THE NEXT STEERING COMMITTEE**
>
> Technical safeguards — value limits, audit logs, anomaly alerts — are necessary and not sufficient. Before granting any system autonomy, ask the three philosophical ones: is the hierarchy of purposes documented, including what happens when objectives conflict? Does it know when it does not know, and escalate? Can it recognise a situation it has never seen, or does it only optimise inside the world it was trained on? Without those three, autonomy is risk wearing the costume of efficiency.

FOUR

Sources

- Schrage, M., & Kiron, D. (2025). Philosophy eats AI. *MIT Sloan Management Review*. The three pillars — teleology, epistemology and ontology — and the argument that the binding constraint on value from AI is philosophical rather than technical, are theirs.
- Schrage, M., & Kiron, D. (2024). Is philosophy the next LLM training frontier? *MIT Initiative on the Digital Economy*.
- Lima, T. F. (2026). *Excelência operacional: da estratégia à execução*, ch. 5. Atena Editora. Open access. The treatment of the three pillars for operations, the executive dilemmas, and the predictive-to-agentic comparison.
- Van Mieghem, J. A., & Allon, G. (2015). *Operations Strategy: Principles and Practice* (2nd ed.). Dynamic Ideas.
- Lima, T. F. (2026). *Excelência operacional*, ch. 1, for the identity V = C × (A + P).

The mapping between the three pillars and the operational arguments of the session — teleology to order winners, epistemology to signalling at the source, ontology to causal ambiguity — is this course's, not the original authors'.

International Certificate in Project Management and Organizational Performance
Module 3 — Methods, Tools, and Operational Efficiency · Digital Tool Appraisal
Sorbonne Business School · IDGP Academy · Prof. Dr. Thiago Lima